

COURSE NAME & CODE: ECA 0303 – Signals and systems

NAME: D S Lakshmi Narayana

REG NO: 192512371

Experiment 3.2

Energy and Power Calculation of Given Signals

Aim:

Calculate energy and power for given signals.

Program:

% Energy and Power Calculation of Signals

clc;

clear;

close all;

% Time vector

t = -10:0.001:10;

dt = t(2) - t(1);

%% Energy Signal

x1 = exp(-2 * abs(t));

% Energy

E1 = sum(abs(x1).^2) * dt;

% Average Power

```
P1 = E1 / (max(t) - min(t));
```

```
%% Power Signal
```

```
x2 = cos(2 * pi * 5 * t);
```

```
% Energy
```

```
E2 = sum(abs(x2).^2) * dt;
```

```
% Average Power
```

```
P2 = E2 / (max(t) - min(t));
```

```
%% Display Results
```

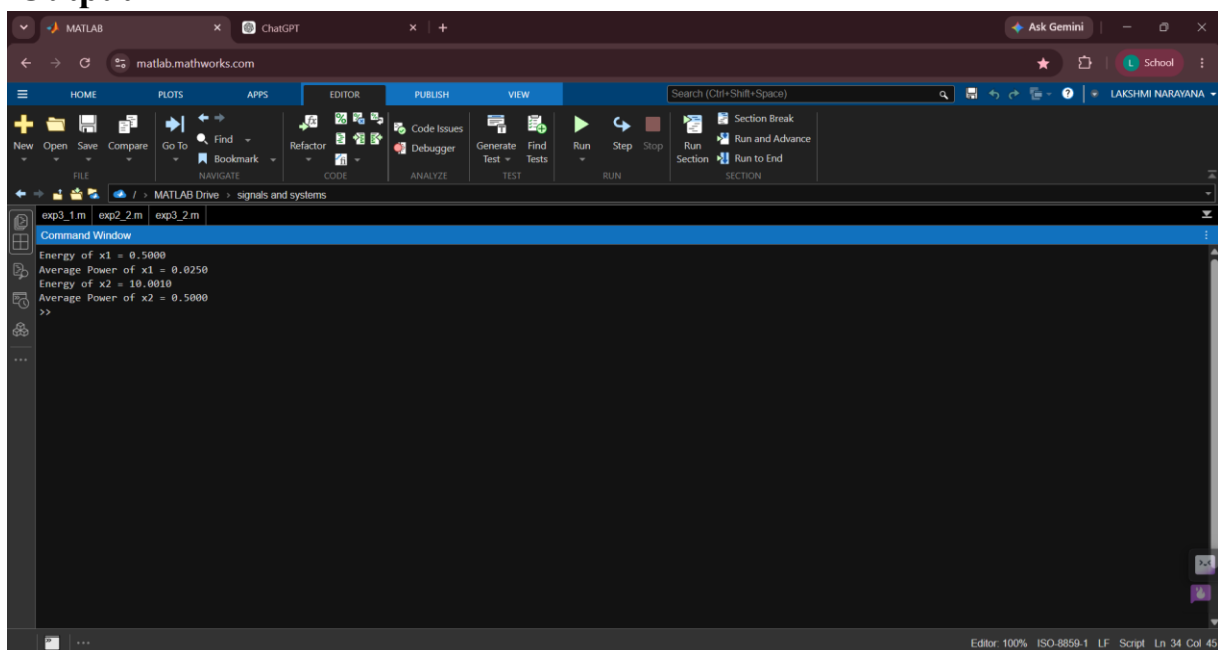
```
fprintf('Energy of x1 = %.4f\n', E1);
```

```
fprintf('Average Power of x1 = %.4f\n', P1);
```

```
fprintf('Energy of x2 = %.4f\n', E2);
```

```
fprintf('Average Power of x2 = %.4f\n', P2);
```

Output



Result:

$x_1 \rightarrow$ Energy signal, $x_2 \rightarrow$ Power signal.